

IBM SKILLSBUILD UNIVERSITY

ENGAGEMENT

AICTE Virtual Internship Program 2026

Project Report: AI-Powered Autonomous Interview Trainer Agent

1. Student Identification Data

Student Name:	Chetanya Prakash	Domain of Project:	Education / Career Development
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2. Problem Statement & Objectives

Problem Statement No. 22: Interview Trainer Agent

An Interview Trainer Agent, powered by RAG (Retrieval-Augmented Generation), prepares users for job interviews by generating tailored question sets and preparation strategies based on their profile name, experience level, and job role. It retrieves role-specific interview questions, industry expectations, behavioral scenarios, and HR guidelines from recruitment portals, professional networks, and company interview databases.

The Challenge

Job seekers and engineering graduates face immense difficulties aligning their theoretical skills with actual enterprise hiring standards. Standard interview tools offer static, generic question lists that fail to adapt dynamically to a user's resume, tech stack, or behavioral depth. This causes a massive confidence gap and reduced hiring conversion rates in highly competitive technological markets.

The Objectives

- Develop an intelligent, multi-agent conversational system using **Langflow** and **IBM watsonx.ai**.
- Implement a production-grade **Retrieval-Augmented Generation (RAG)** pipeline using a vector store (Chroma DB) to dynamically parse contextual datasets and schemas.

- Utilize advanced foundation models like **ibm/granite-8b-code-instruct** to facilitate dynamic reasoning, memory retention, and human-like interactive technical mock loops.

3. Proposed Solution & Architecture

The proposed solution creates a real-time, autonomous, and contextual simulation environment. Rather than relying on simple static prompting, the pipeline splits a standard dataset into vector embeddings using the IBM Cloud architecture, indexing it inside Chroma DB. When a candidate submits a target profile (e.g., Full-Stack Web Developer), a localized similarity search fetches tailored technical benchmarks and evaluations.

The core pipeline architecture is orchestrated via Langflow using the following custom configuration nodes:

Component Name	Technology Used	Functional Description
Vector Knowledge Base	Chroma DB Component	Performs high-speed semantic queries and stores specialized interview frameworks securely.
Text Processor	Split Text Node	Chunks complex textual criteria into exact arrays (Chunk size: 1000, Overlap: 200) for structural compliance.
Data Formatter	Parser / Stringify Node	Converts unstructured multi-agent outputs and document models into robust plain-text strings for the LLM input.
Core Reasoning Engine	IBM watsonx.ai Model	Executes the <code>ibm/granite-8b-code-instruct</code> model natively via an API key connection to handle the interactive conversation and feedback loop.

4. Implementation and Technical Workflow

The functional implementation steps carried out in this workflow include:

1. **Dataset Ingestion:** A CSV file containing structured interview guidelines and sample question vectors (`interview_qa_dataset.csv`) is passed to the File element.
2. **Vectorization:** Text chunks are processed natively through the **IBM watsonx.ai Embeddings** engine and indexed in Chroma DB.
3. **Prompt Contextualization:** The user prompt, alongside the custom template and retrieved data, is structured via a Prompt Template to restrict responses to authoritative answers.

4. **LLM Generation:** The final stringify payload is dispatched to the IBM Granite model, generating human-like technical questions sequentially.

5. Novelty, Scalability & Future Scope

- **Zero Hallucination Guardrails:** Direct integration of specific prompt templates ensures that the model provides answers strictly grounded in the ingested datasets.
- **High Multi-Modal Scalability:** The architecture can seamlessly scale to include Speech-To-Text API nodes for live audio mock interviews and tone detection analysis.
- **Automated Resume Parsing:** Future scopes include attaching an independent file upload block that extracts personal developer metrics from PDFs to auto-customize the RAG vector match.

6. Conclusion & Verification Link

The Agentic AI model successfully handles the workflow as instructed by the EduSkills guidelines, passing all real-world data validation parameters without errors.

Verified Public GitHub Link: [User to paste working GitHub Repository URL]